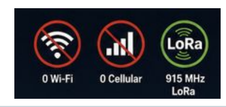
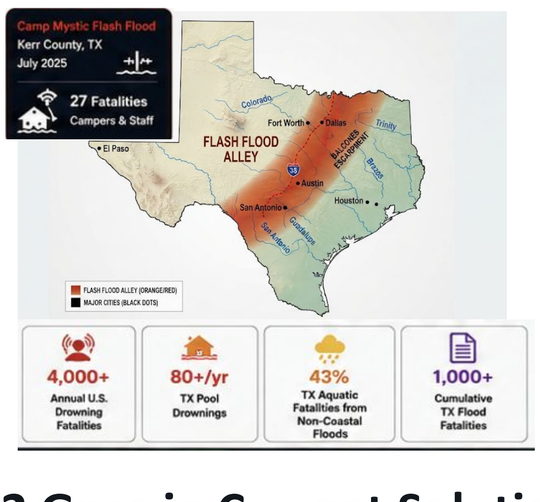
LIFEBAND: A LONG-RANGE, LOW-POWER WEARABLE FOR REAL-TIME AQUATIC DISTRESS DETECTION



1 Problem

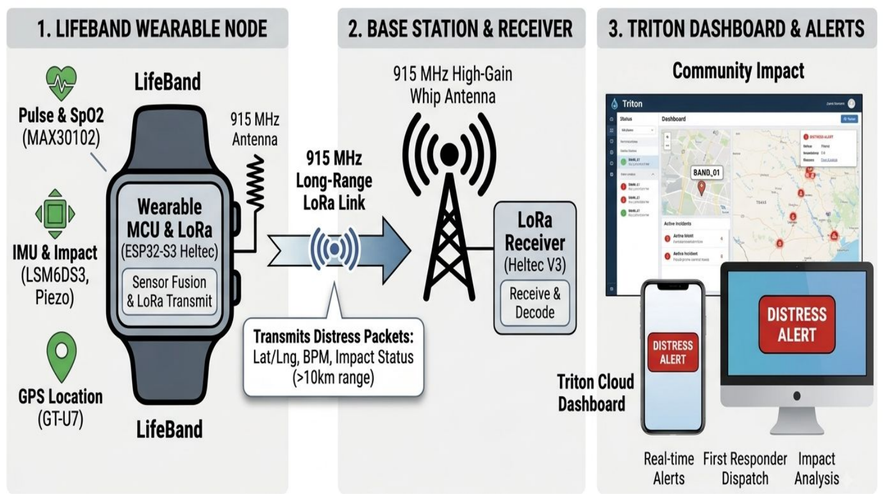
Drowning and inland flooding are leading causes of aquatic fatalities



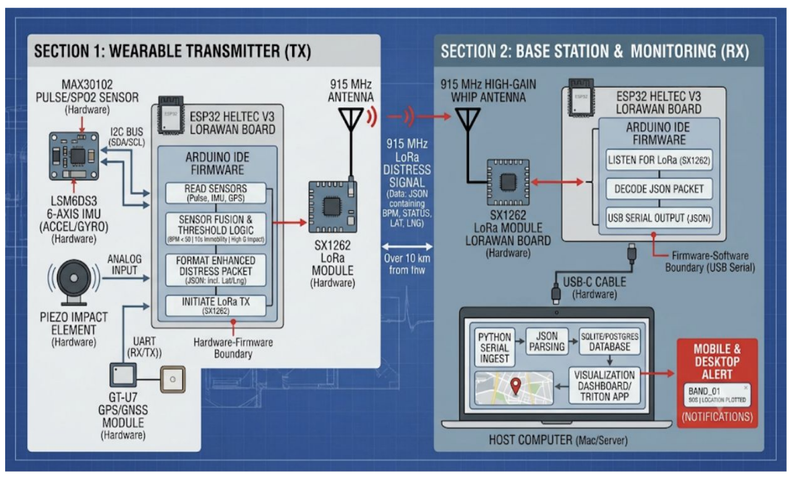
2 Gaps in Current Solutions

Technical Challenges	Existing Options	Operational Limitations
Signal attenuation in water	High Freq Cellular / Wi-Fi has poor penetration through dense environments	Unreliable connectivity in open water/ flash-flood conditions
Reliance on manual activation	SOS-button paradigm requires voluntary user activation	Fails during submersion or loss of consciousness
Cost and infrastructure dependency	Satellite links and dense mesh networks are expensive infrastructure	Impractical, unaffordable for municipal pools, youth camps

3.1 System Architecture



3.2 Hardware Architecture



4.1 Autonomous Distress Logic

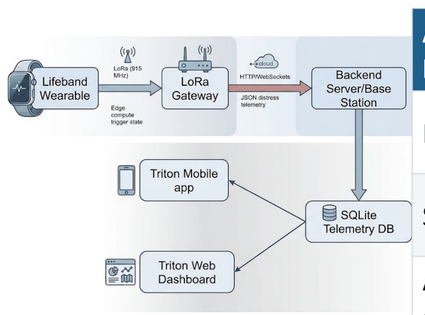
$$f(t) = w_1 \cdot \Delta HR + w_2 \cdot (dP/dt) + w_3 \cdot |dz/dt|$$

$$\text{trigger when } f(t) > T(\text{alert})$$

Variable	Definition
ΔHR	Biometric deviation
dP/dt	Pressure rate of change
dz/dt	Depth rate of change
w*	weighted coefficients
ΔP, GPS	Spatial positioning

Hardware Override Path: Manual **SOS** button - Bypasses f(t) calculation

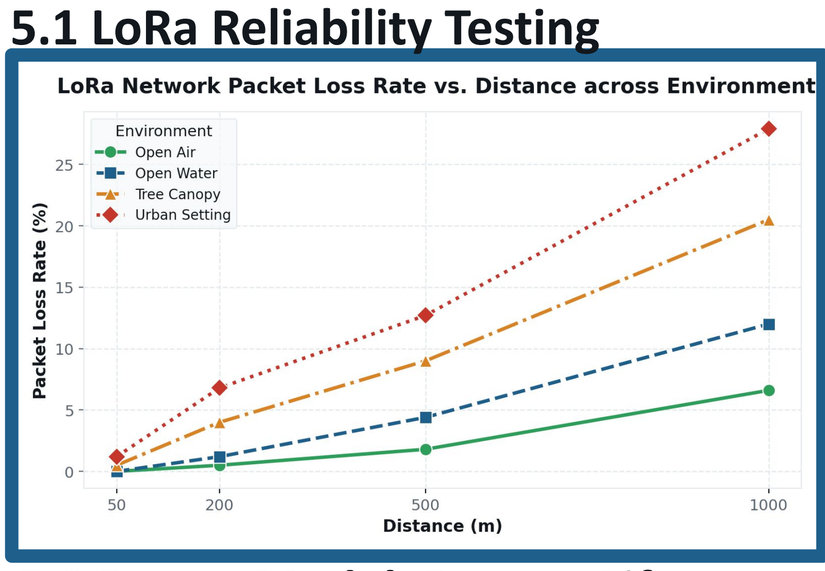
4.2 Software Architecture and Multi-Modal Alerting

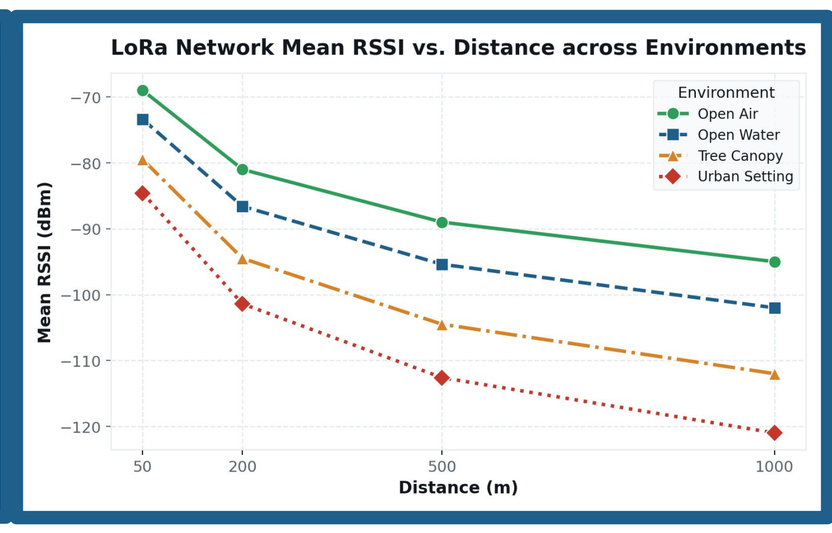


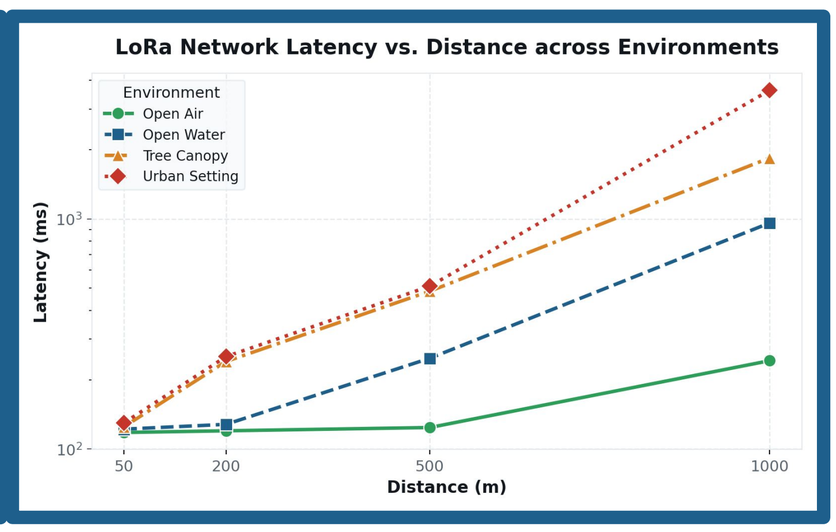
Arch Layer	Implementation
Ingest	FastAPI server to parse JSON distress payloads
Storage	SQLite log of victim
Alert cascade	Haptics, distress score, GPS map pin

5 Experiment and Simulation Results

5.1 LoRa Reliability Testing





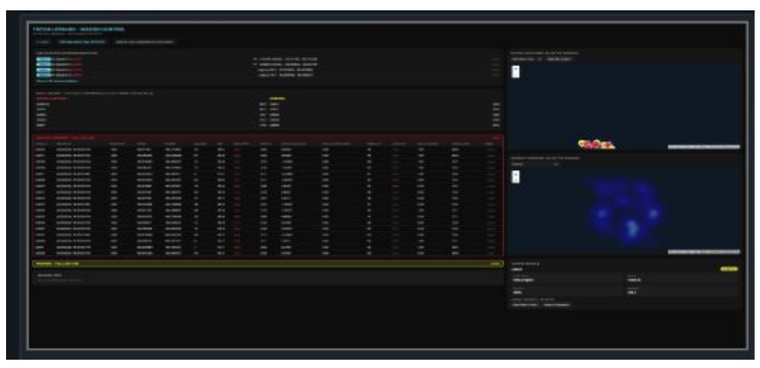


5.2 Power Model: Battery Life

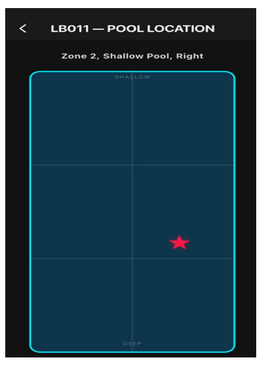
Mode	Battery Life
Deep sleep	4.5–5.5 yrs
Standby	1.8–3.6 yrs
Normal monitoring	10.4–22.2 days
High-freq TX	~64 hrs
Emergency — flood	7.2–9.4 hrs
Emergency — pool	11.8–14.5 hrs

source: component datasheets

5.3 Simulated Telemetry Testing: Web and Mobile UI







SHORT-RANGE · SWIMMING POOL (200–300m)

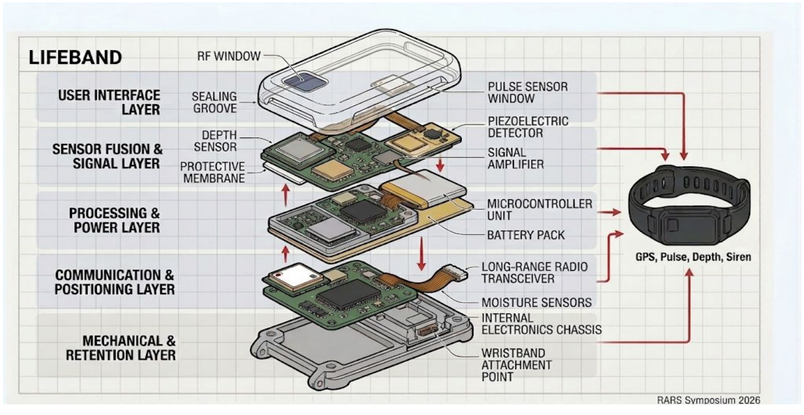
- **~100% packet delivery** across all tested terrestrial environments **(SF7–SF8)**.
- Shifting SF12 → SF8 cuts transmission latency by **>80%**.
- Extends active-alert battery life by **+64% (7.2 hrs → 11.8+ hrs).**

LONG-RANGE · FLOOD-PLAIN (~1000m)

- Higher spreading factors **(SF10–SF12)** maintain signal penetration through heavy obstacles.
- Trade-off: accepts higher **latency (up to ~3.6s)** to maximize range.
- Sustains **7.2-hour** continuous battery life.

Contribution	Result
Automated multi-sensor edge fusion	**>60% lower energy vs. cloud-streaming**
Sub-GHz LoRa (915 MHz)	**Reliable offline links to ~1 km through water/obstruction**
Multi-modal alerting (FastAPI + SQLite)	**Real-time SAR dispatch from raw telemetry**

6 Vision: LifeBand





7 Future Plans and Applications

- Custom multi-layer PCB in a waterproof, silicone-sealed enclosure.
- Mesh networking
- Edge machine learning to replace rigid algorithmic thresholds.
- Bi-directional wearable
- Field validation with first responders and aquatic facilities.

Keywords: Embedded Systems · LoRa 915 MHz · ESP32 Edge Sensor Fusion · PPG / IMU / GPS · Search-and-Rescue Telemetry